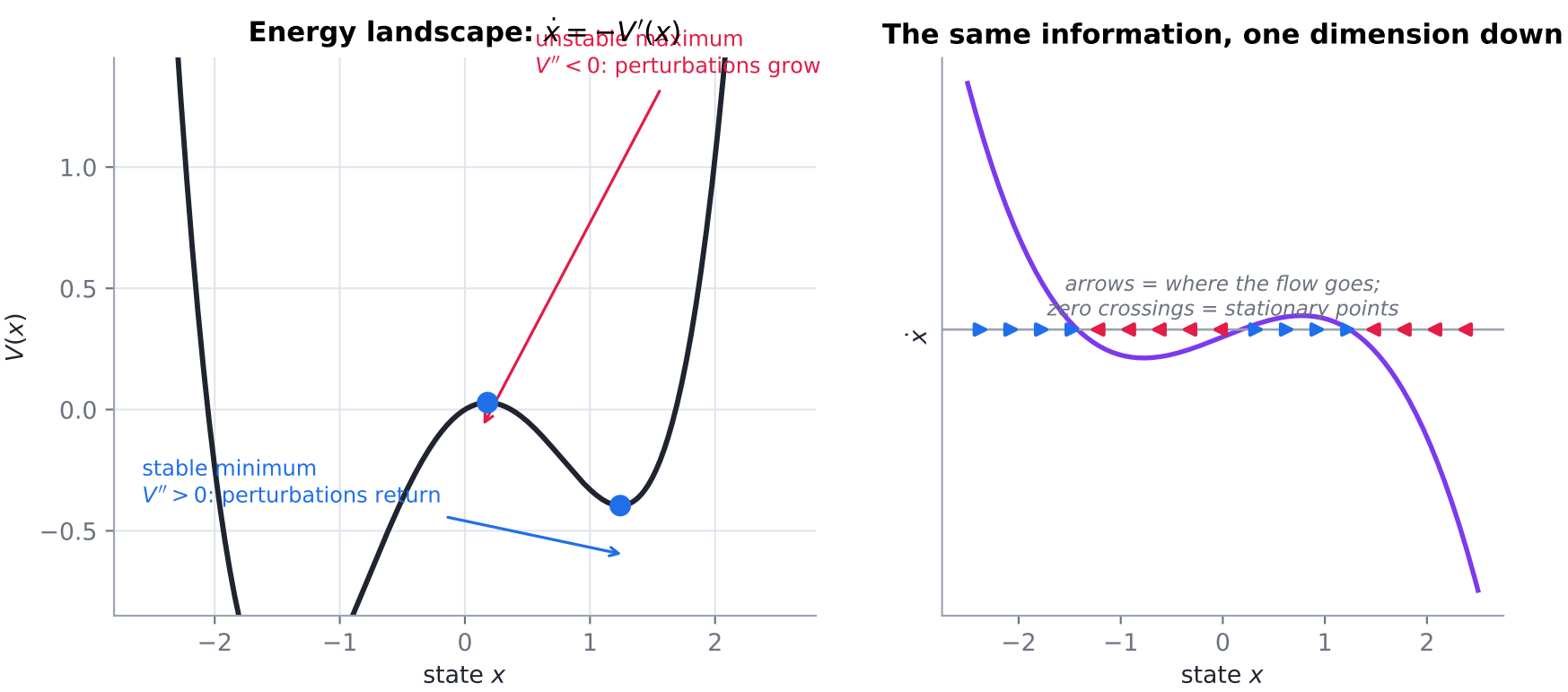
Energy landscape: $\dot{x} = -V'(x)$
unstable maximum
$V'' < 0$: perturbations grow
stable minimum
$V'' > 0$: perturbations return
$V(x)$
1.0
0.5
0.0
−0.5
−2
−1
0
1
2
state x
The same information, one dimension down
arrows = where the flow goes;
zero crossings = stationary points
$\dot{x}$
−2
−1
0
1
2
state x